Stable item 2376

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Editorial problem boundary. Prove precisely the original Theorem1.1 below: count integer coefficient triples of height at most B that admit a unit resultant with some odd-degree integer polynomial, and obtain the printed B-cubed/square-root-log bound. Retain the full resultant-minor elimination, explicit rational conic obstruction and prime-square local sieve count. The selected route needs only rational X,Y for the nonzero coefficients and the i=1,j=2 conic used in the primary, not a universal polynomial-coefficient assertion for every minor choice. Fixed-degree torsion heuristics, discriminant ordering and later improvements are unselected context.

Difficulty (editorial): H2. A unit resultant with an unspecified odd degree does not immediately give a coefficientwise sieve restriction independent of that degree. The authors eliminate Sylvester columns and evaluate minors to extract a rational quadratic conic whose solvability forces a prime-square discriminant/nonresidue obstruction. Computing the exclusion density1/(2p) from that new reduction supplies the nonroutine arithmetic step; the precise cited large-sieve and multiplicative estimates then produce the logarithmic saving.

Literal source scope note. The original polynomial-coefficient wording in Lemmas2.1/2.2 is preserved. The selected primary explicitly asks only for rational X, Y and uses its $i = 1, j = 2$ conic; no polynomial-versus-rational-function correction is supplied.

Editorial transcription note. Original author language, mathematics and ordering are preserved. Publisher front matter is replaced by this independent article wrapper. Bibliography is recovered from matching cached publisher HTML, including its TeX formulas. No new proof or mathematical repair is supplied. Surrounding original results are retained for conservative context and are not extra selected corpus claims.

1 Introduction

Let $Q \in \mathbb{Z}[x]$ be a quadratic polynomial and let $R \in \mathbb{Z}[x]$ be a polynomial of degree n. Associated to Q and R is the resultant $\mathrm{Res}(R, Q)$. This is a polynomial of degree $n+2$ in the coefficients of Q and R, which is defined over $\mathbb{Z}$. Given any fixed polynomial $R \in \mathbb{Z}[x]$ of degree $n \geq 5$, it follows from work of Schmidt [16, Th. 1] that there are only finitely many irreducible quadratic polynomials $Q \in \mathbb{Z}[x]$ such that $\mathrm{Res}(R, Q) = \pm 1$. In this paper we shall concern ourselves with the opposite situation, and attempt to assess the density of quadratic $Q \in \mathbb{Z}[x]$ for which $\mathrm{Res}(R, Q) = \pm 1$, for some $R \in \mathbb{Z}[x]$ of odd degree. More specifically, we shall be interested in the size of the counting function

$$N(B) = \#\left\{ (a,b,c) \in \mathbb{Z}^3 : \begin{array}{l} |a|, |b|, |c| \leq B \\ \mathrm{Res}(R, ax^2+bx+c) = \pm 1 \\ \text{for some } R \in \mathbb{Z}[x] \text{ of odd degree} \end{array} \right\}, \tag{1.1}$$

as $B \to \infty$, with the aim of showing that $N(B) = o(B^3)$.

By using only mod p information, we shall apply the large sieve to prove the following upper bound for $N(B)$.

Theorem 1.1. *We have $N(B) \ll B^3/\sqrt{\log B}$.*

It is also interesting to ask about the behaviour of the function

$$N_n(B) = \#\left\{ (a,b,c) \in \mathbb{Z}^3 : \begin{array}{l} |a|, |b|, |c| \leq B \\ \mathrm{Res}(R, ax^2+bx+c) = \pm 1 \\ \text{for some } R \in \mathbb{Z}[x] \text{ of degree } n \end{array} \right\}, \tag{1.2}$$

for fixed $n \in \mathbb{N}$. Theorem 1.1 upper bounds $N_n(B)$ when n is odd. Taking $R(x) = x^n$, it is easily checked that $\mathrm{Res}(x^n, ax^2+bx+c) = c^n$. Hence we have $N_n(B) \gg B^2$, coming from triples (a,b,c) with $c = 1$. In private communication with the authors, Aaron Landesman has raised the following conjecture.

Conjecture 1.2 (Landesman)**.** *Let $n \geq 2$. Then there exists $m \geq 0$ such that*

$$N_n(B) \ll B^2 (\log B)^m.$$

One might even expect that this upper bound holds with $m = 0$ (at least when n is odd). We shall give some evidence towards Conjecture 1.2, by relating it to standard expectations around the typical size of n-torsion in the class group of imaginary quadratic fields.

Suppose that we are given a monic polynomial $R \in \mathbb{Z}[x]$ of odd degree, and a prime p. Then, as is well-known, a quadratic polynomial $Q \in \mathbb{Z}[x]$ will share a common root in $\overline{\mathbb{F}}_p$ with R if and only if $p \mid \mathrm{Res}(R, Q)$. It follows that the quadratic polynomials $Q \in \mathbb{Z}[x]$ that share no common root with R in $\overline{\mathbb{F}}_p$ for any prime p, are precisely the quadratic polynomials $Q \in \mathbb{Z}[x]$ for which $\mathrm{Res}(R, Q) = \pm 1$. On appealing to Theorem 1.1, we may conclude as follows.

Corollary 1.3. *Let $R \in \mathbb{Z}[x]$ be monic and of odd degree. Then almost all quadratic polynomials $Q \in \mathbb{Z}[x]$ share a common root with R in $\overline{\mathbb{F}}_p$, for some prime p.*

Elements of the class group of a quadratic number field $K/\mathbb{Q}$ of discriminant D_K are in bijection with equivalence classes of binary quadratic forms of discriminant D_K. This can be used to handle the average size of the class number $h(K)$, if one orders the quadratic number fields by discriminant. Thus it follows from Dirichlet's class number formula that

$$\sum_{\substack{K/\mathbb{Q} \text{ quadratic} \\ |D_K| < X}} h(K) \ll X^{3/2}. \tag{1.3}$$

It is much more challenging to assess the typical size of $h_{\mathrm{odd}}(K)$, which is defined to be the odd part of the class number, or even that of $h_n(K)$, for fixed n, which is defined to be size of the n-torsion subgroup of the class group. In his thesis work [15], Landesman provided a new geometric description of the n-torsion elements of the class groups of quadratic number fields, for a fixed n. Thus it follows from [15, Th. 1.1] that a binary quadratic form q corresponds to an odd order element in the class group if and only if there exists an odd integer n and a degree n polynomial whose resultant with q is ± 1. Combining this with the proof of Theorem 1.1, we will deduce the following result in Sections 5.1 and 5.2.

Corollary 1.4. *We have*

$$\sum_{\substack{K/\mathbb{Q} \text{ quadratic} \\ |D_K|<X}} h_{\mathrm{odd}}(K) \ll \frac{X^{3/2}}{\sqrt{\log X}}.$$

It is also possible to arrive at this upper bound using Gauss' genus theory, which yields the lower bound $h(K) = h_{\mathrm{odd}}(K)h_{2^\infty}(K) \geq h_{\mathrm{odd}}(K)2^{\omega(D_K)-1}$ for the class number. But then, on restricting to negative discriminants $D_K < -4$, it follows from Dirichlet's class number formula that

$$\sum_{\substack{K/\mathbb{Q} \text{ quadratic} \\ -X<D_K<-4}} h_{\mathrm{odd}}(K) \leq \frac{2}{\pi} \sum_{\substack{K/\mathbb{Q} \text{ quadratic} \\ -X<D_K<-4}} \frac{\sqrt{|D_K|}\, L(1,\chi)}{2^{\omega(D_K)}}, \tag{1.4}$$

where $L(s,\chi)$ is the Dirichlet L-function with primitive real character $\chi(\bullet) = (\frac{D_K}{\bullet})$. The sum on the right hand side can be estimated in a variety of ways, but recent work of Wilson [17, Cor. 1.6] yields an upper bound for (1.4) that matches the one in Corollary 1.4. For real quadratic fields $K/\mathbb{Q}$ one can argue similarly, but this time taking the trivial lower bound $R_K \gg \log|D_K|$ for the regulator of K in Dirichlet's class number formula. In this way it is possible to obtain the improvement

$$\sum_{\substack{K/\mathbb{Q} \text{ quadratic} \\ 0<D_K<X}} h_{\mathrm{odd}}(K) \ll \sum_{\substack{K/\mathbb{Q} \text{ quadratic} \\ 0<D_K<X}} \frac{\sqrt{D_K}\, L(1,\chi)}{(\log D_K)\, 2^{\omega(D_K)}} \ll \frac{X^{3/2}}{(\log X)^{3/2}}.$$

Gerth's [10, 11] extension of the Cohen–Lenstra heuristics to the 2-part of the class group implies that the upper bound in Corollary 1.4 is tight.

For a fixed positive integer n, there are many results in the literature that provide upper bounds for the average of $h_n(K)$ that save a power of X, rather than a power of $\log X$. After significant contributions by Heath-Brown–Pierce [12], by Ellenberg–Pierce–Wood [5], and by Frei–Widmer [8], the best general bound is currently

$$\sum_{\substack{K/\mathbb{Q} \text{ quadratic} \\ |D_K|<X}} h_n(K) \ll X^{3/2-1/(n+1)+\varepsilon},$$

for any $\varepsilon > 0$, and is due to Koymans–Thorner [14]. Note that any argument that yields a power-saving over the trivial bound for $N_n(B)$ ought to yield a corresponding power-saving for the average of $h_n(K)$. In Section 3 we shall discuss some of the issues involved when $n = 3$.

As a special case of the Cohen–Lenstra heuristics [4], it is widely believed that

$$\sum_{\substack{K/\mathbb{Q} \text{ quadratic} \\ |D_K|<X}} h_n(K) \ll \begin{cases} X & \text{if } n \text{ is odd,} \\ X\log X & \text{if } n \text{ is even.} \end{cases} \tag{1.5}$$

It turns out that by using Landesman's correspondence, we can apply this bound to deduce an upper bound for the variant $N_n^-(B)$ of $N_n(B)$ in (1.2), in which a restriction to quadratics with negative discriminants is imposed. We shall prove the following result in Section 5.1, the second part of which gives strong evidence towards Conjecture 1.2.

Theorem 1.5. *1. We have $N_n^-(B) \ll_\varepsilon B^{3-2/(n+1)+\varepsilon}$, for any $\varepsilon > 0$.*

2. Assume that (1.5) *holds. Then $N_n^-(B) \ll B^2(\log B)^2$.*

Unfortunately it seems difficult to establish similar upper bounds for the counting function $N_n(B)$, since for quadratics with positive discriminant we are required to estimate the number of distinct integers with absolute value up to x that are represented by a binary quadratic form with positive discriminant close to x^2, and such a bound would inevitably depend on the size of the fundamental unit.

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2 Resultants

Let $Q(x) = ax^2+bx+c$ be a quadratic polynomial with $a, b, c \in \mathbb{Z}$, and let $R(x) = r_0x^n+r_1x^{n-1}+\cdots+r_n$ be a polynomial of degree n also defined over $\mathbb{Z}$. The resultant of R and Q is defined to be the determinant of the $(n+2) \times (n+2)$ matrix M, where

$$M = \begin{pmatrix} r_0 & 0 & a & 0 & \cdots & 0 & 0 \\ r_1 & r_0 & b & a & \cdots & 0 & 0 \\ r_2 & r_1 & c & b & \cdots & 0 & 0 \\ r_3 & r_2 & 0 & c & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \\ r_n & r_{n-1} & 0 & 0 & \cdots & c & 0 \\ 0 & r_n & 0 & 0 & \cdots & 0 & c \end{pmatrix}.$$

In this section our main goal is to prove a formula for $\mathrm{Res}(R, Q) = \det M$ which is tailored to the problem of estimating $N_n(B)$ in (1.2). For any positive integer k, let A_k be the $(k+2) \times k$ matrix

$$A_k = \begin{pmatrix} a & 0 & \cdots & 0 \\ b & a & \cdots & 0 \\ c & b & \cdots & 0 \\ 0 & c & \cdots & a \\ \vdots & \vdots & \ddots & b \\ 0 & 0 & \cdots & c \end{pmatrix}. \tag{2.1}$$

In what follows, we shall denote by $B_{k,i,j}$ the $k \times k$ matrix formed by removing the i-th and j-th row from A_k, for any $1 \le i \le k+2$ and $1 \le j \le k$. With the notation (2.1), if $\boldsymbol{c} = (r_0, \ldots, r_n)$, then the first two columns of M are given by $(\boldsymbol{c}, 0)^T$ and $(0, \boldsymbol{c})^T$, respectively, and the remaining n columns are given by A_n.

We begin by using elementary row operations to clear all but two of the entries from the first two columns of the matrix whose determinant is being taken.

Lemma 2.1. *Let $1 \le i < j \le n+1$. Then there exist $X, Y \in \mathbb{Q}[r_0, \ldots, r_n, a, b, c]$ depending on the choice of i, j such that*

$$\mathrm{Res}(R, Q) = \det \begin{pmatrix} \vdots & \vdots & \\ X & 0 & \\ 0 & X & \\ \vdots & \vdots & A_n \\ Y & 0 & \\ 0 & Y & \\ \vdots & \vdots & \end{pmatrix},$$

where X appears in the $(i, 1)$-entry and $(i+1, 2)$-entry of the matrix, and Y appears in the $(j, 1)$-entry and $(j+1, 2)$-entry.

Proof. Let i, j be fixed, with $1 \le i < j \le n+1$. Let $\boldsymbol{c}$ be the column vector $(r_0, \ldots, r_n)^T$ and let $\boldsymbol{c}_{i,j}$ denote the column vector formed by removing the i-th and j-th entries from $\boldsymbol{c}$. Recalling the definition (2.1) of A_k, and the subsequent definition of $B_{k,i,j}$, we claim that we can find a linear combination over

$\mathbb{Q}$ of the columns of A_n to clear any $n-1$ of the $n+1$ non-trivial entries of the first column. Indeed, we begin by solving

$$B_{n-1,i,j}\boldsymbol{\lambda} = \boldsymbol{c}_{i,j},$$

for a vector $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_{n-1})^T \in \mathbb{Q}^{n-1}$. We then proceed to perform the column operation which subtracts $A_n(\lambda_1, \dots, \lambda_{n-1}, 0)^T$ from the first column of M, or $A_n(0, \lambda_1, \dots, \lambda_{n-1})^T$ from the second column of M. The $n+2$ rows of A_n are given by

$$\begin{aligned} \boldsymbol{a}_1 &= (a, 0, \dots, 0), & \boldsymbol{a}_2 &= (b, a, \dots, 0), \\ \boldsymbol{a}_{n+1} &= (0, \dots, 0, c, b), & \boldsymbol{a}_{n+2} &= (0, \dots, 0, c), \end{aligned}$$

and

$$\boldsymbol{a}_k = (0, \dots, 0, c, b, a, 0, \dots, 0),$$

for $3 \le k \le n$, where c, b, a occur in the $k-2, k-1, k$-th entry, respectively. In conclusion, the procedure we have described produces $X, Y \in \mathbb{Q}[r_0, \dots, r_n, a, b, c]$, given by

$$\begin{aligned} X &= r_{i-1} - \boldsymbol{a}_i \begin{pmatrix} \boldsymbol{\lambda} \\ 0 \end{pmatrix} = r_{i-1} - c\lambda_{i-2}\mathbf{1}_{i\ge 3} - b\lambda_{i-1}\mathbf{1}_{i\ge 2} - a\lambda_i \mathbf{1}_{i\le n-1}, \\ Y &= r_{j-1} - \boldsymbol{a}_j \begin{pmatrix} \boldsymbol{\lambda} \\ 0 \end{pmatrix} = r_{j-1} - c\lambda_{j-2}\mathbf{1}_{j\ge 3} - b\lambda_{j-1}\mathbf{1}_{j\le n} - a\lambda_j \mathbf{1}_{j\le n-1}, \end{aligned}$$

such that the statement of the lemma holds. □

Using this expression for the resultant we are now in a position to compute the determinant.

Lemma 2.2. *Let $1 \le i < j \le n+1$. Then there exist $X, Y \in \mathbb{Q}[r_0, \dots, r_n, a, b, c]$ depending on the choice of i, j such that*

$$\operatorname{Res}(R, Q) = a^{i-1}c^{n-j+1}\big(c^{j-i}X^2 + (-1)^{j-i}(d_{j-i} - acd_{j-i-2})XY + a^{j-i}Y^2\big),$$

where $d_{-1} = 0$, $d_0 = 1$, and d_k is the $k \times k$ determinant

$$d_k = \det \begin{pmatrix} b & a & 0 & \cdots & & & \\ c & b & a & \cdots & & 0 & \\ 0 & c & b & \cdots & & & \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ & & & \dots & b & a & 0 \\ & 0 & & \dots & c & b & a \\ & & & \dots & 0 & c & b \end{pmatrix},$$

for $k \in \mathbb{N}$.

Proof. Recall the definition (2.1) of A_k, and the subsequent definition of $B_{k,i,j}$. (In particular $d_k = \det B_{k,1,k+2}$ in this notation, for $k \ge 1$.) Let i, j be fixed such that $1 \le i < j \le n+1$. Then it follows from Lemma 2.1 that

$$\operatorname{Res}(R, Q) = (-1)^{i+1}X \det M_{i,1} + (-1)^{j+1}Y \det M_{j,1},$$

where $M_{k,1}$ is the $(n+1) \times (n+1)$ matrix by removing the k-th row and the first column. Now the matrix $M_{i,1}$ has an X in its $(i,1)$-entry and a Y in its $(j,1)$-entry, so

$$\det M_{i,1} = (-1)^{i+1}X \det B_{n,i,i+1} + (-1)^{j+1}Y \det B_{n,i,j+1}.$$

Similarly, the matrix $M_{j,1}$ has an X in its $(i+1,1)$-entry if $j > i+1$, and a Y in its $(j,1)$-entry, so

$$\det M_{j,1} = (-1)^{i+2}X\mathbf{1}_{j>i+1} \det B_{n,i+1,j} + (-1)^{j+1}Y \det B_{n,j,j+1}.$$

Putting everything together, we obtain

$$\mathrm{Res}(R,Q) = X^2 \det B_{n,i,i+1} + Y^2 \det B_{n,j,j+1} + (-1)^{j-i} (\det B_{n,i,j+1} - \mathbf{1}_{j>i+1} \det B_{n,i+1,j})\, XY. \quad (2.2)$$

It remains to compute $\det B_{n,i,j+1}$ for integers $1 \le i \le j \le n+1$. If $i > 1$, Laplace expansion along the first row of $B_{n,i,j+1}$ gives

$$\det B_{n,i,j+1} = a \det B_{n-1,i-1,j}.$$

If $j \le n$, Laplace expansion along the final row of $B_{n,i,j+1}$ gives

$$\det B_{n,i,j+1} = c \det B_{n-1,i,j+1}.$$

Applying these relations recursively, we easily obtain

$$\det B_{n,i,j+1} = a^{i-1} \det B_{n-i+1,1,j-i+2} = a^{i-1} c^{n-j+1} \det B_{j-i,1,j-i+2}.$$

Recalling the definition of d_k from the statement, we see that

$$\det B_{n,i,j+1} = a^{i-1} c^{n-j+1} d_{j-i},$$

which easily leads to the statement of the lemma, on insertion into (2.2). □

Each d_k is a homogeneous polynomial of degree k in a, b, c and can be evaluated as follows.

Lemma 2.3. *Assume that $b^2 - 4ac \neq 0$. Then for any $k \in \mathbb{N}$, we have*

$$d_k = \frac{1}{\sqrt{b^2-4ac}} \left(\left(\frac{b+\sqrt{b^2-4ac}}{2} \right)^{k+1} - \left(\frac{b-\sqrt{b^2-4ac}}{2} \right)^{k+1} \right).$$

Furthermore, for $k \ge 2$, we have

$$d_k - acd_{k-2} = \left(\frac{b+\sqrt{b^2-4ac}}{2} \right)^{k} + \left(\frac{b-\sqrt{b^2-4ac}}{2} \right)^{k}.$$

Proof. This follows easily from the recurrence relation $d_k = bd_{k-1} - acd_{k-2}$, and the initial conditions $d_1 = b$ and $d_2 = b^2 - ac$. □

3 The cubic case: a worked example

In this section we discuss the counting function $N_3(B)$ in (1.2), which we shall use to illustrate the calculations in the preceding section. When $R \in \mathbb{Z}[x]$ is cubic we have

$$\begin{aligned} \mathrm{Res}(R,Q) &= \det \begin{pmatrix} r_0 & 0 & a & 0 & 0 \\ r_1 & r_0 & b & a & 0 \\ r_2 & r_1 & c & b & a \\ r_3 & r_2 & 0 & c & b \\ 0 & r_3 & 0 & 0 & c \end{pmatrix} \\ &= a^3 r_3^2 - a^2 b r_2 r_3 - 2a^2 c r_1 r_3 + a^2 c r_2^2 + a b^2 r_1 r_3 + 3abc r_0 r_3 \\ &\quad - abc r_1 r_2 - 2ac^2 r_0 r_2 + ac^2 r_1^2 - b^3 r_0 r_3 + b^2 c r_0 r_2 - bc^2 r_0 r_1 + c^3 r_0^2. \end{aligned}$$

The points $(a, b, c, r_0, r_1, r_2, r_3)$ on the hypersurface $\mathrm{Res}(R,Q) = -1$ are in bijection with the points $(-a, -b, -c, r_0, r_1, r_2, r_3)$ on the hypersurface $\mathrm{Res}(R,Q) = 1$, and so we may focus our attention on the latter equation. This defines a quintic hypersurface in $\mathbb{A}^7 = \mathbb{A}^3_{a,b,c} \times \mathbb{A}^4_{r_0,\dots,r_3}$ and has the structure of a quadric bundle over $\mathbb{A}^3_{a,b,c}$, the fibres of which are quadrics in $\mathbb{A}^4_{r_0,\dots,r_3}$. In this way we can interpret $N_3(B)$ as counting integral points of height at most B in $\mathbb{A}^3_{a,b,c}$ for which the fibre has an integral point. However, it turns out that these quadrics are all degenerate and we are led to study the existence of integral points in families of irreducible affine conics.

We can get explicit equations for the relevant conics by revisiting the calculations in Lemma 2.1. The set $\{(1,2),(1,3),(1,4),(2,3),(2,4),(3,4)\}$ comprises the set of allowable indices (i,j), so that we always have $j-i\in\{1,2,3\}$. If $(i,j)=(1,2)$, then

$$\mathrm{Res}(R,Q)=\det\begin{pmatrix}X_{1,2}&0&a&0&0\\Y_{1,2}&X_{1,2}&b&a&0\\0&Y_{1,2}&c&b&a\\0&0&0&c&b\\0&0&0&0&c\end{pmatrix}=c^2(cX_{1,2}^2-bX_{1,2}Y_{1,2}+aY_{1,2}^2),$$

where

$$\begin{pmatrix}X_{1,2}\\Y_{1,2}\end{pmatrix}=\begin{pmatrix}r_0\\r_1\end{pmatrix}-\begin{pmatrix}a&0\\b&a\end{pmatrix}\begin{pmatrix}\lambda_1\\\lambda_2\end{pmatrix}\quad\text{and}\quad\begin{pmatrix}c&b\\0&c\end{pmatrix}\begin{pmatrix}\lambda_1\\\lambda_2\end{pmatrix}=\begin{pmatrix}r_2\\r_3\end{pmatrix}.$$

If $(i,j)=(1,3)$, then $\mathrm{Res}(R,Q)$ is

$$\det\begin{pmatrix}X_{1,3}&0&a&0&0\\0&X_{1,3}&b&a&0\\Y_{1,3}&0&c&b&a\\0&Y_{1,3}&0&c&b\\0&0&0&0&c\end{pmatrix}=c(c^2X_{1,3}^2+(b^2-2ac)X_{1,3}Y_{1,3}+a^2Y_{1,3}^2),$$

where

$$\begin{pmatrix}X_{1,3}\\Y_{1,3}\end{pmatrix}=\begin{pmatrix}r_0\\r_1\end{pmatrix}-\begin{pmatrix}a&0\\c&b\end{pmatrix}\begin{pmatrix}\lambda_1\\\lambda_2\end{pmatrix}\quad\text{and}\quad\begin{pmatrix}b&a\\0&c\end{pmatrix}\begin{pmatrix}\lambda_1\\\lambda_2\end{pmatrix}=\begin{pmatrix}r_2\\r_3\end{pmatrix}.$$

If $(i,j)=(1,4)$, then $\mathrm{Res}(R,Q)$ is

$$\det\begin{pmatrix}X_{1,4}&0&a&0&0\\0&X_{1,4}&b&a&0\\0&0&c&b&a\\Y_{1,4}&0&0&c&b\\0&Y_{1,4}&0&0&c\end{pmatrix}=c^3X_{1,4}^2-(b^3-3abc)X_{1,4}Y_{1,4}+a^3Y_{1,4}^2,$$

where

$$\begin{pmatrix}X_{1,4}\\Y_{1,4}\end{pmatrix}=\begin{pmatrix}r_0\\r_1\end{pmatrix}-\begin{pmatrix}a&0\\0&c\end{pmatrix}\begin{pmatrix}\lambda_1\\\lambda_2\end{pmatrix}\quad\text{and}\quad\begin{pmatrix}b&a\\c&b\end{pmatrix}\begin{pmatrix}\lambda_1\\\lambda_2\end{pmatrix}=\begin{pmatrix}r_2\\r_3\end{pmatrix}.$$

These calculations are all consistent with Lemmas 2.2 and 2.3.

When one makes all these substitutions for $X_{i,j},Y_{i,j}$ one see that $N_3(B)$ is bounded by the number of $(a,b,c)\in\mathbb{Z}^3$ for which $|a|,|b|,|c|\leq B$ and the equations

$$\begin{aligned}cx_2^2-bx_2y_2+ay_2^2&=c^2,\\c^2x_3^2+(b^2-2ac)x_3y_3+a^2y_3^2&=b^2c,\\c^3x_4^2-(b^3-3abc)x_4y_4+a^3y_4^2&=(b^2-ac)^2,\end{aligned}$$

each admit a solution $(x_j,y_j)\in\mathbb{Z}^2$, for $2\leq j\leq 4$. Testing for solubility of generalised Pell equations presents a formidable challenge, as exemplified in recent work [3, 6, 7, 13] on the solubility of the negative Pell equation.

4 The large sieve

We now turn to the task of estimating the counting function $N(B)$ that was defined in (1.1). Note that there are $O(B^2)$ choices of (a,b,c) with $abc=0$. Moreover, the resultant $\mathrm{Res}(R,ax^2+bx+c)$ is homogeneous of odd degree in the coefficients a,b,c. Hence, on replacing (a,b,c) by $(-a,-b,-c)$, we have

$$N(B)\leq 2\#\left\{(a,b,c)\in\mathbb{Z}^3:\ \begin{array}{l}0<|a|,|b|,|c|\leq B\\ \mathrm{Res}(R,ax^2+bx+c)=1\\ \text{for some } R\in\mathbb{Z}[x] \text{ of odd degree}\end{array}\right\}+O(B^2).$$

We now suppose that $\mathrm{Res}(R, ax^2+bx+c) = 1$ for some $R \in \mathbb{Z}[x]$ of odd degree $n \geq 3$. Then, for any choice of indices (i,j) with $1 \leq i < j \leq n+1$, it follows from Lemma 2.2 that there exist $X, Y \in \mathbb{Q}$ such that

$$a^{i-1}c^{n-j+1}\left(c^{j-i}X^2 + (-1)^{j-i}(d_{j-i} - acd_{j-i-2})XY + a^{j-i}Y^2\right) = 1,$$

where formulae for d_{j-i} and d_{j-i-2} are given by Lemma 2.3 in terms of a, b, c. Letting $k = j-i$, we may conclude that the conic $C_{i,j} \subset \mathbb{P}^2$ has a rational point, where $C_{i,j}$ is given by

$$c^kX^2 + (-1)^k(d_k - acd_{k-2})XY + a^kY^2 = a^{i-1}c^{n-j+1}Z^2.$$

Unfortunately, there is no advantage to be gained from examining the existence of rational points on more than one conic. Taking $(i,j) = (1,2)$ and making an obvious change of variables, we see that $C_{1,2}(\mathbb{Q}) \neq \emptyset$ if and only if the conic

$$C: \quad X^2 - (b^2-4ac)Y^2 = cZ^2$$

has a $\mathbb{Q}$-point. It is worth highlighting that this is the key step at which our assumption on the parity of the degree of R enters the argument. We conclude that

$$N(B) \leq 2N_+(B) + O(B^2), \tag{4.1}$$

with

$$N_+(B) = \#\big\{(a,b,c) \in \mathbb{Z}^3 : 0 < |a|, |b|, |c| \leq B,\ C(\mathbb{Q}) \neq \emptyset\big\}.$$

If $C(\mathbb{Q}) \neq \emptyset$ then necessarily $(\frac{c}{p}) \neq -1$ for any prime p such that $v_p(b^2-4ac) = 1$. Thus

$$N_+(B) \leq T(B,B,B), \tag{4.2}$$

where if $B_1, B_2, B_3 \geq 1$ we put

$$T(B_1,B_2,B_3) = \#\big\{(a,b,c) \in \Omega : |a| \leq B_1,\ |b| \leq B_2,\ |c| \leq B_3\big\},$$

where

$$\Omega = \left\{(a,b,c) \in \mathbb{Z}^3 : v_p(b^2-4ac) = 1 \Rightarrow \left(\frac{c}{p}\right) \neq -1 \text{ for all primes } p\right\}.$$

We shall approach the problem of estimating the right hand side via the large sieve, in the form [1, Lem. 6.3]. We first record an upper bound for the size of the reduction modulo p^2 of Ω.

Lemma 4.1. *Then* $\#(\Omega \bmod p^2) \leq (1-\omega_p)p^6$, *where* $\omega_p = 1/2p + O(1/p^2)$.

Proof. We may clearly assume that $p > 2$. We have $\#(\Omega \bmod p^2) = p^6 - S_p$, where S_p is the number of $(a,b,c) \in (\mathbb{Z}/p^2\mathbb{Z})^3$ for which $v_p(b^2-4ac) = 1$ and $(\frac{c}{p}) = -1$. Clearly

$$S_p \geq p^3 \#\left\{(a,b,c) \in \mathbb{F}_p^3 : b^2-4ac = 0,\ \left(\frac{c}{p}\right) = -1\right\} - \#\big\{(a,b,c) \in (\mathbb{Z}/p^2\mathbb{Z})^3 : b^2-4ac \equiv 0 \bmod p^2\big\}.$$

The first summand is $\frac{1}{2}p^4(p-1)$ and the second summand is $O(p^4)$. Hence we deduce that $S_p \geq \frac{1}{2}p^5 + O(p^4)$, which concludes the proof. □

It now follows from [1, Lem. 6.3] that

$$T(B_1,B_2,B_3) \ll \frac{(B_1+Q^4)(B_2+Q^4)(B_3+Q^4)}{L(Q)},$$

for any $Q \geq 1$, where

$$L(Q) = \sum_{1 \leq q \leq Q} \mu^2(q) \prod_{p \mid q} \frac{\omega_p}{1-\omega_p} \geq \sum_{1 \leq q \leq Q} \frac{\mu^2(q)}{\tau(q)q} \prod_{p \mid q} \Big(1 - \frac{1}{p}\Big)^C$$

for an appropriate absolute constant $C \geq 1$. A straightforward application of [9, Th. A.5] now yields $L(Q) \gg \sqrt{\log Q}$, from which it follows that

$$T(B_1,B_2,B_3) \ll \frac{B_1B_2B_3}{\sqrt{\log \min\{B_1,B_2,B_3\}}}, \tag{4.3}$$

on taking $Q = \min\{B_1,B_2,B_3\}^{1/4}$. Theorem 1.1 now follows on taking $B_1 = B_2 = B_3 = B$ and inserting this into (4.1) and (4.2).

5 Ordering by discriminant

5.1 Negative discriminant

Let

$$S^-(X) = \sum_{\substack{K/\mathbb{Q} \text{ quadratic} \\ -X<d<0}} h_{\mathrm{odd}}(K),$$

where $d = D_K$ is the discriminant of K. In this section we shall compare $S^-(X)$ with variants of the counting function

$$N^-(B) = \#\left\{(a,b,c) \in \mathbb{Z}^3 : \begin{array}{l} |a|,|b|,|c| \le B,\ b^2 < 4ac \\ \mathrm{Res}(R, ax^2+bx+c) = \pm 1 \\ \text{for some } R \in \mathbb{Z}[x] \text{ of odd degree} \end{array}\right\}, \tag{5.1}$$

in which a restriction to negative discriminants is imposed in the counting function (1.1). We begin by proving the following result, which takes care of the negative discriminants in Corollary 1.4.

Lemma 5.1. *We have*

$$S^-(X) \ll \frac{X^{3/2}}{\sqrt{\log X}}.$$

Proof. As described by Buell [2, Prop. 6.20], there is an isomorphism between the set of equivalence classes of primitive binary quadratic forms of discriminant d and the narrow ideal class group of K. A quadratic form $q(x,y) = ax^2+bxy+cy^2$ with discriminant $d<0$ is *reduced* (in the sense of [2, Eq. (2.1)]) if

$$|b| \le a \le c.$$

Any binary quadratic form with discriminant d is equivalent to a reduced quadratic form of discriminant d by [2, Th. 2.3]. In the light of the work by Landesman [15, Th. 1.1], the sum $S^-(X)$ is bounded by the number of reduced primitive positive definite quadratic forms q of discriminant $d \in (-X, 0)$ for which there exists polynomial $R \in \mathbb{Z}[x]$ of odd degree whose resultant with $q(x,1)$ is ± 1. If $|d| = 4ac - b^2 \ge 3a^2$, so that $a \ll \sqrt{d} \ll \sqrt{X}$. If $|b| \le X^{1/2-1/10}$, then $|d| \le X$ implies that $ac \ll X$. Dropping the resultant condition, we see that the overall contribution to $S^-(X)$ from such binary quadratic forms is $O(X^{3/2-1/10}\log X)$, which is satisfactory. Therefore we can proceed under the assumption that $|b| > X^{1/2-1/10}$. Since $b^2 \le a^2 \ll |d|$, we have $ac \ll |d| + b^2 \ll X$. We therefore concentrate on the set of $(a,b,c) \in \mathbb{Z}_{\ge 0} \times \mathbb{Z} \times \mathbb{Z}_{\ge 0}$ which are constrained by the inequalities

$$X^{1/2-1/10} < |b| \le a \ll \sqrt{X} \quad \text{and} \quad a \le c \ll \frac{1}{a}X.$$

It will be convenient to sort the contribution into dyadic intervals, with

$$A \le a < 2A, \quad B \le |b| < 2B, \quad C \le c < 2C, \tag{5.2}$$

where A, B, C run over powers of 2 and satisfy

$$X^{1/2-1/10} \ll B \ll A \ll \sqrt{X} \quad \text{and} \quad A \ll C \ll \frac{1}{A}X. \tag{5.3}$$

Let $S^-(A,B,C)$ denote the overall contribution to $S^-(X)$ from the binary quadratic forms $ax^2+bxy+cy^2$ whose coefficients are constrained by (5.2). Then it follows from (4.3) that

$$S^-(A,B,C) \ll \frac{ABC}{\sqrt{\log B}} \ll \frac{ABC}{\sqrt{\log X}},$$

since $C \gg A \gg B \gg X^{1/2-1/10}$. It remains to sum this estimate for A, B, C running over powers of 2 that are constrained by the inequalities in (5.3). This yields

$$\begin{aligned} S^-(X) \leq \sum_{A,B,C} S^-(A,B,C) &\ll \sum_{A,C} \frac{A^2 C}{\sqrt{\log X}} \\ &\ll \frac{X}{\sqrt{\log X}} \sum_{A \ll \sqrt{X}} A \\ &\ll \frac{X^{\frac{3}{2}}}{\sqrt{\log X}}, \end{aligned}$$

which thereby completes the proof of the lemma. □

The latter result shows how an upper bound for a variant of the counting function (5.1), in which lop-sided box sizes are allowed, can be used to provide an upper bound for $S^-(X)$. Fixing a choice of integer $n \in \mathbb{N}$, our next result reverses the process and shows how $N_n^-(B)$ can be bounded in terms of the sum

$$S_n^-(X) = \sum_{\substack{K/\mathbb{Q} \text{ quadratic} \\ -X<d<0}} h_n(K).$$

Lemma 5.2. *Let $n \in \mathbb{N}$. Then*

$$N_n^-(B) \ll B^2 \sum_{j=0}^{2\log_2 B + O(1)} \frac{S_n^-(2^j)}{2^j}.$$

Proof. Note that we can trivially impose that $\gcd(a,b,c) = 1$, because we always have $\gcd(a,b,c)^n \mid \mathrm{Res}(R, ax^2+bx+c)$ for any $R \in \mathbb{Z}[x]$ with degree n. The discriminant of $q(x,y) = ax^2+bxy+cy^2$ can be written as Dr^2, where D is a fundamental discriminant and $r \in \mathbb{Z}$ is non-zero. In particular $|D| \leq |D|r^2 \ll B^2$, since $\max\{|a|,|b|,|c|\} \leq B$.

By [2, Prop. 7.1], there exists a form f of discriminant D, and $\alpha, \beta, \gamma, \delta \in \mathbb{Z}$, such that $f(\alpha x+\beta y, \gamma x+\delta y) = q(x,y)$ and $\alpha\delta - \beta\gamma = r$. The condition $|a|, |c| \leq B$ implies that

$$|f(\alpha,\gamma)| = |a| \leq B, \quad \text{and} \quad |f(\beta,\delta)| = |c| \leq B. \tag{5.4}$$

Without loss of generality, we can take $f(x,y) = a_0x^2 + b_0xy + c_0y^2$ to be a reduced form, so that in particular $0 \leq a_0 \ll \sqrt{|D|} \ll B$. Since $D = b_0^2 - 4a_0c_0 < 0$, we must have $a_0c_0 \neq 0$. Thus, on writing

$$f(x,y) = \frac{1}{4a_0}\big((2a_0x + b_0y)^2 + |D|y^2\big),$$

we see that $|f(x,y)| \leq B$ implies that $|y| \leq \sqrt{4a_0B/|D|}$ and $|x + (b_0/2a_0)y| \leq \sqrt{B/a_0}$. Therefore

$$\#\{(x,y) \in \mathbb{Z}^2 : |f(x,y)| \leq B,\ y \neq 0\} \ll \sqrt{\frac{4a_0B}{|D|}}\left(\sqrt{\frac{B}{a_0}} + 1\right) \ll \frac{B}{\sqrt{|D|}}.$$

By symmetry, it follows that

$$\#\{(x,y) \in \mathbb{Z}^2 : |f(x,y)| \leq B,\ (x,y) \neq (0,0)\} \ll \frac{B}{\sqrt{|D|}}.$$

Given f, we may conclude that there are $O(B^2/|D|)$ choices of (α,γ) and (β,δ) that satisfy (5.4).

The condition that $\mathrm{Res}(R,q) = \pm 1$ for some $R \in \mathbb{Z}[x]$ of degree n implies that $\mathrm{Res}(R(dx-b, -cx+a), f(x)) = \mathrm{Res}(R(x), f(ax+b, cx+d)) = \pm 1$. Appealing to [15, Th. 1.1], given any fundamental discriminant $D < 1$, we have the upper bound

$$\#\left\{(a,b,c) \in \mathbb{Z}^3 : \begin{array}{l} |a|,|b|,|c| \leq B \\ b^2-4ac = Dr^2 \text{ for some } r \in \mathbb{Z} \\ \mathrm{Res}(R, ax^2+bx+c) = \pm 1 \\ \text{for some } R \in \mathbb{Z}[x] \text{ of degree } n \end{array}\right\} \ll \frac{B^2}{|D|} h_n(\mathbb{Q}(\sqrt{D})).$$

Finally we sum over all D in dyadic intervals up to $D \ll B^2$ to conclude that

$$N_n^-(B) \ll B^2 \sum_{\substack{D<0 \\ |D| \ll B^2}} \frac{h_n(\mathbb{Q}(\sqrt{D}))}{|D|} \ll B^2 \sum_{X=2^j \ll B^2} \frac{S_n^-(X)}{X},$$

as claimed in the lemma. □

Using this result we may easily complete the proof of Theorem 1.5. The second part is a direct consequence of (1.5), since $j = O(\log B)$. For the first part we apply the work of Koymans and Thorner [14], which gives

$$N_n^-(B) \ll_\varepsilon B^2 \sum_{2^j \ll B^2} (2^j)^{1/2-1/(n+1)+\varepsilon} \ll B^{3-2/(n+1)+2\varepsilon},$$

for any $\varepsilon > 0$.

5.2 Positive discriminant

Let

$$S^+(X) = \sum_{\substack{K/\mathbb{Q} \text{ quadratic} \\ 0<d<X}} h_{\mathrm{odd}}(K),$$

where $d = D_K$ is the discriminant of K. In this section we shall prove the following result, which completes the proof of Corollary 1.4, once combined with Lemma 5.1.

Lemma 5.3. *We have*

$$S^+(X) \ll \frac{X^{3/2}}{\sqrt{\log X}}.$$

We shall prove this result by relating $S^+(X)$ to the counting function

$$N^+(B) = \#\left\{(a,b,c) \in \mathbb{Z}^3 : \begin{array}{l} |a|,|b|,|c| \le B,\ b^2 > 4ac \\ \mathrm{Res}(R, ax^2+bx+c) = \pm 1 \\ \text{for some } R \in \mathbb{Z}[x] \text{ of odd degree} \end{array}\right\},$$

in which a restriction to positive discriminants is imposed in (1.1). Thus, since $N^+(B) \le N(B)$, Lemma 5.3 is a direct consequence of applying Theorem 1.1 in the following result.

Lemma 5.4. *We have $S^+(X) \le N^+(\sqrt{X})$.*

Proof. We adapt the arguments in the previous section to deal with quadratic number fields of positive discriminant. A quadratic form $q(x,y) = ax^2 + bxy + cy^2$ with discriminant $d > 0$ is *reduced* (in the sense of [2, Eq. (3.1)]) if

$$0 < b < \sqrt{d}, \quad \sqrt{d} - b < 2|a| < \sqrt{d} + b.$$

Moreover, a reduced form with discriminant $d > 0$ further satisfies

$$\sqrt{d} - b < 2|c| < \sqrt{d} + b,$$

by [2, Prop. 3.1]. Thus

$$|a| < \sqrt{d}, \quad 0 < b < \sqrt{d}, \quad |c| < \sqrt{d},$$

for any reduced binary quadratic form $q(x,y)$ with discriminant $d > 0$.

Any binary quadratic form with discriminant d is equivalent to a reduced quadratic form of discriminant d by [2, Prop. 3.3]. It follows from the work of Landesman [15, Th. 1.1] that the sum $S^+(X)$ is bounded by the number of reduced primitive positive definite quadratic forms q of discriminant $d \in (0, X)$, for which there exists an odd degree polynomial $R \in \mathbb{Z}[x]$ whose resultant with $q(x,1)$ is ± 1. Hence, in order to bound $S^+(X)$, we can restrict to triples $(a,b,c) \in \mathbb{Z}$ in the range $\max\{|a|,|b|,|c|\} \le \sqrt{X}$ and the statement easily follows. □

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